Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

Team name: TL · Taivanbat Bayarsaikhan (leader), Lkhagvadorj Munkhdalai

AI Engineers & Data Scientists at M bank, Mongolia's first neobank (fully digital, licensed 2022)

Problem Statement: BFSI — Intelligent Risk, Fraud & Financial Experiences. Kervan Credit Passport: an AI underwriting agent that turns a small business's informal records (ledger photos, bank statements, payment exports) into a verified, explainable credit passport a bank can act on. Demo market: Mongolia.

M bank

Brief about the idea

The problem

- Most Mongolian micro and small businesses keep their books in a handwritten ledger, a bank account and a QR payment app. Banks cannot underwrite that at scale, so thin-file businesses are refused or overpriced.
- Declared figures are unverifiable, manual review is slow, and decisions are opaque to the owner.

Kervan Credit Passport

- The owner uploads a ledger photo, a bank statement (PDF/CSV) and a payment export, and declares revenue, tenure, amount and tenor.
- In about a minute the agent returns a credit passport (grade K1–K7 with an indicative PD, rate band and suggested limit from Kervan's v0.1 policy scorecard, to be recalibrated on partner-bank data), an owner explanation in English and Mongolian with a re-scored improvement plan, and an underwriter-ready credit memo with an evidence appendix.
- Core principle: Gemini reads, verifies, explains and writes; points, penalties and grade come only from a deterministic, auditable scorecard. Gemini never computes a score and can never raise one: at most it tightens an input's verification status, and its flags carry no penalty.
- Banks receive anonymized passports and make indicative offers; identity is revealed only when the business accepts, and the bank's own KYC and credit approval follow before anything is disbursed.

Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

How it differs

• Verification, not extraction: declared vs documented and documents vs each other (ledger vs bank, payments vs bank, totals vs re-summed rows, account holder, tenure vs account age); arithmetic in Python; every figure traced to a page or row.

• Explainable by construction: a points scorecard with reason codes; self-declared inputs count half, inconsistent inputs count zero.

• Reads the records people actually have: handwritten Mongolian ledger photos, local bank CSV/PDF exports, payment settlements.

How it solves the problem · USP

• Banks: an underwriter-ready memo and evidence trail in a minute, inconsistencies flagged; every memo is a draft for a human underwriter (K1–K2 need sign-off, K3–K6 are referred, K7 gets no standard offer).

• Owners: a plain-language grade in Mongolian and an improvement plan with gains computed by re-running the scorer.

• USP: trust by design — number guardrails, no protected attributes, cross-source consistency checks today (authenticity and registration documents on the roadmap), identity disclosure controlled by the business.

List of features offered by the solution

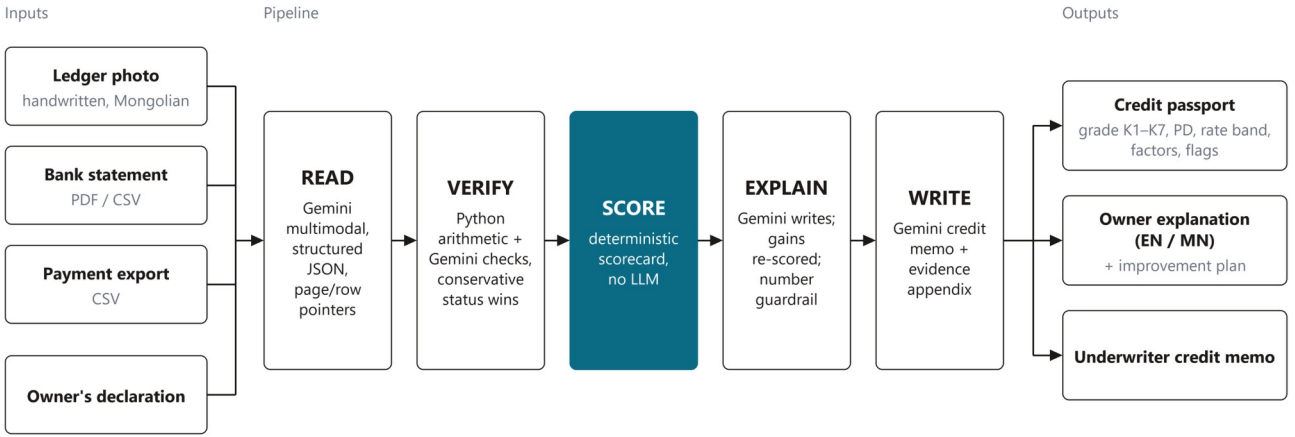
Agent pipeline (live, streamed to the UI)

- READ: Gemini multimodal, structured JSON; monthly figures with document + page/row pointers.
- VERIFY: Python re-sums CSVs and checks continuity, volatility, declared-vs-bank, ledger-vs-bank, payments-vs-bank, account age; Gemini adds qualitative checks; the more conservative status wins.
- SCORE: deterministic scorecard, no LLM; per-factor points, reason codes, penalties.
- EXPLAIN: owner summary in English and Mongolian; improvement plan with re-scored gains.
- WRITE: credit memo with cash-flow analysis, risks, mitigants, scorecard table and evidence appendix.

Product views

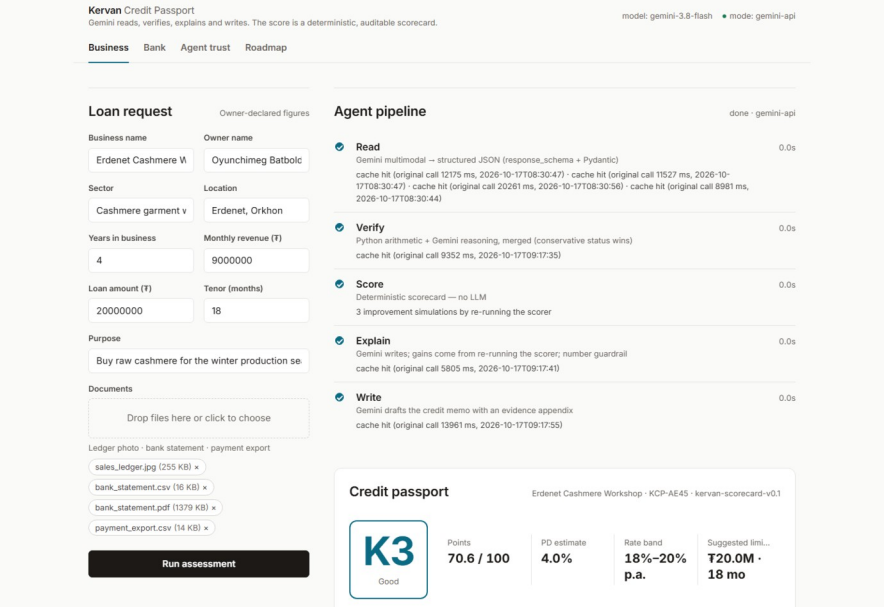
- Business: loan request, uploads, live pipeline, passport card with status-encoded factor bars, flags, EN/MN toggle, improvement plan, offers.
- Bank: anonymized passport list, flags, memo, make-offer; identity revealed only when the business accepts.
- Agent trust: guardrails, step trace (prompt version, model, latency, schema validation, cache), scorer inputs, verification checks, evidence trail with links to the documents.

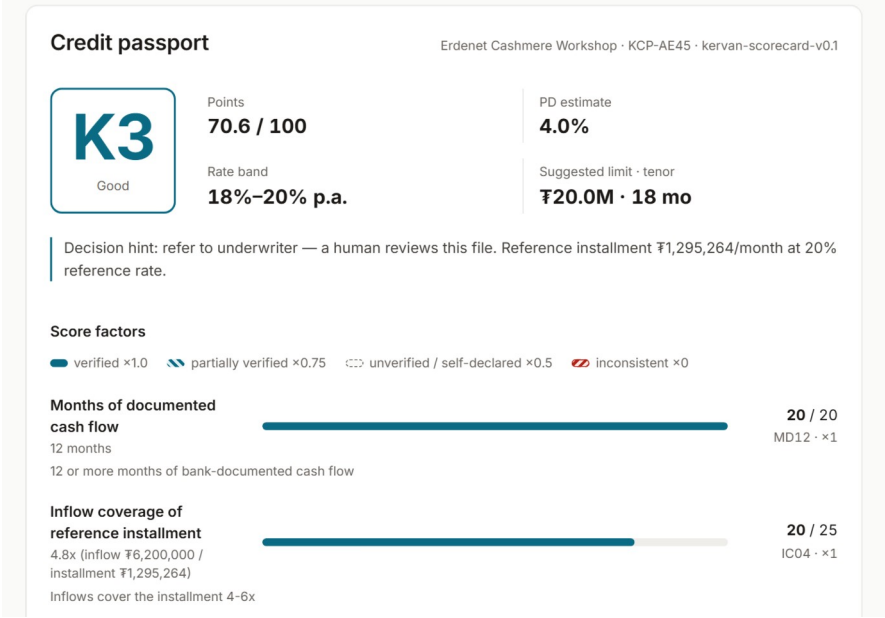
Process flow diagram or Use-case diagram



Gemini reads, verifies, explains and writes. The score comes only from the scorecard.

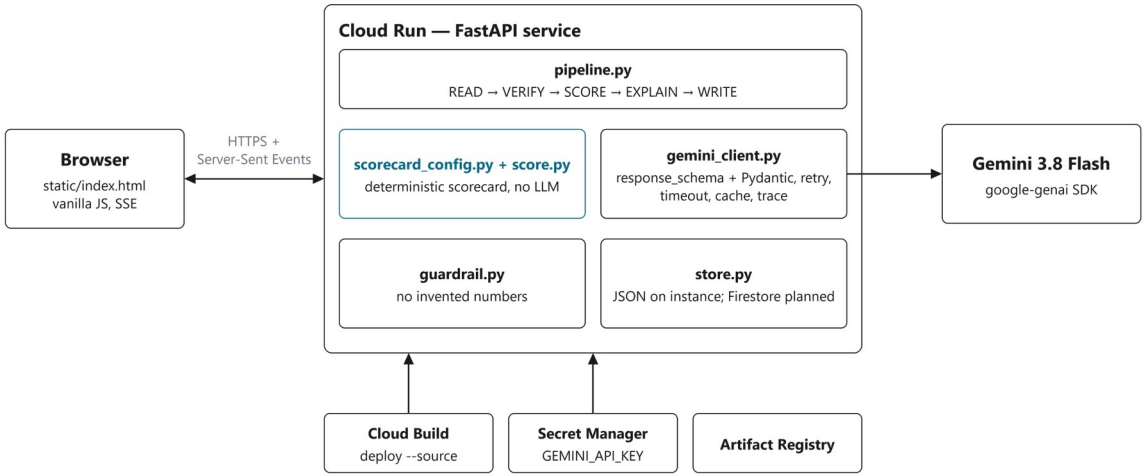
Wireframes/Mock diagrams of the proposed solution (optional)





Left: loan request and live agent pipeline. Right: the credit passport with status-encoded factor bars, flags and the re-scored improvement plan.

Architecture diagram of the proposed solution



Browser (static HTML/JS served by FastAPI) → Cloud Run service → Gemini 3.8 Flash via google-genai (structured JSON, one call per document in parallel). Build: Cloud Build from source → Artifact Registry. Secrets: Secret Manager → env at runtime. Progress streams back over Server-Sent Events.

Technologies to be used in the solution

Google Cloud and AI

- Gemini 3.8 Flash via the google-genai SDK: JSON output with a Pydantic response_schema on all four model steps; low thinking level for sub-minute runs; one retry feeding the validation error back; per-call timeout; cache keyed by model, prompt version and file hash. The same code runs on Vertex AI with the Cloud Run service account (no API key).
- Cloud Run: deploy from source (Cloud Build → Artifact Registry), 1 vCPU / 1 GiB, scales to zero. Secret Manager holds the Gemini key; the runtime service account has only secretAccessor; no secret in the repo or image.
- Storage: JSON in /tmp on the Cloud Run instance for the demo (resets on cold start); Firestore persistence for passports, offers and traces is the planned next step.
- Access: the demo service is public and holds synthetic data only; Cloud Run IAM or Identity-Aware Proxy gates it before real customer documents.

Application stack

- Python 3.12, FastAPI, Pydantic v2; Server-Sent Events for live progress.
- Deterministic verification math and a points scorecard in one config file; unit tests pin the behaviour.
- Number guardrail: every figure in generated text must exist in the inputs; retry once, then deterministic fallback.
- Front-end: one static page, vanilla HTML/CSS/JS.
- Prompts versioned as text files with the guardrails inside.

Estimated implementation cost (optional)

Per assessment (prototype measurements)

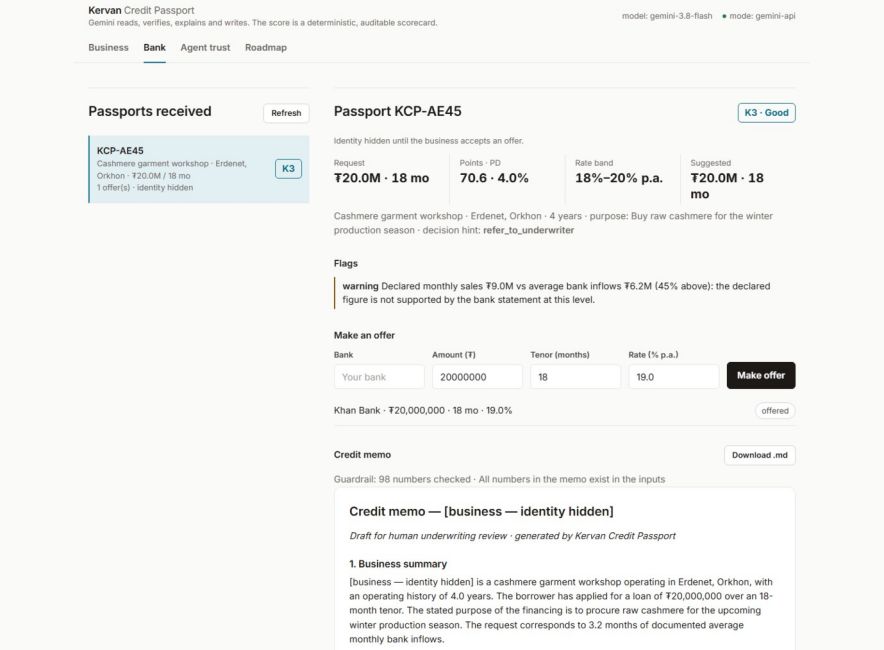
- 4 to 7 Gemini Flash calls per assessment (one per document plus verify, explain, write); 60–120k input tokens; about US$0.02–0.05 at Flash list prices. Repeated runs on the same files are served from the cache at zero cost.
- Cloud Run with scale-to-zero: compute only while a request runs (~1 vCPU-minute per assessment); within the free tier for pilot volumes.

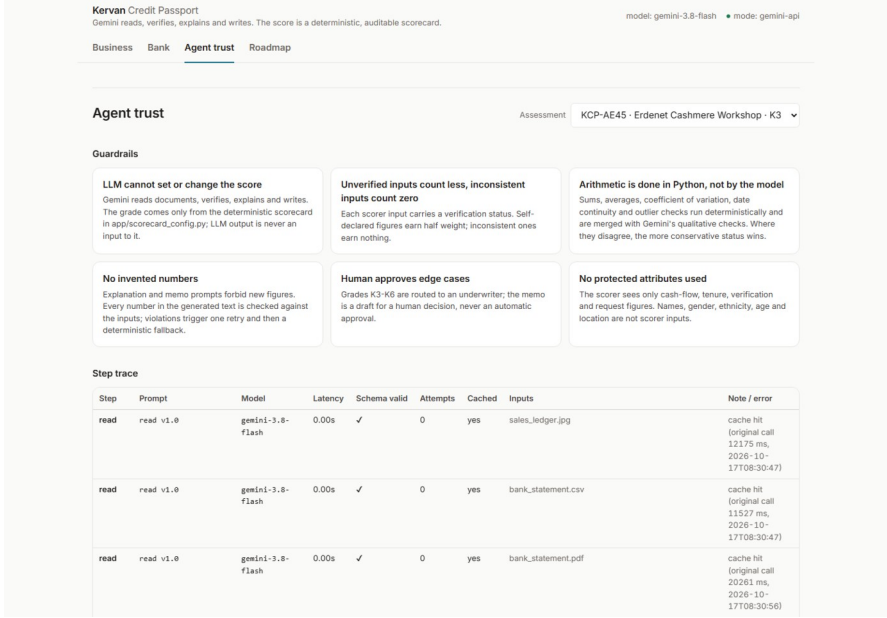
Monthly, indicative

- Pilot (1,000 assessments / month): ≈ US$30–60 Gemini + ≈ US$5–15 Cloud Run + Secret Manager/Artifact Registry cents.
- Scale (50,000 assessments / month): ≈ US$1.5–2.5k Gemini + ≈ US$150–300 Cloud Run; Firestore adds cents per thousand operations.
- No per-seat licences; the scorecard and verification run in plain Python.

Estimates use public list prices in October 2026 and the prototype's measured token volumes; actual cost depends on document sizes.

Snapshots of the prototype





Left: Bank view — anonymized passport, offer, Gemini-drafted credit memo. Right: Agent trust — guardrails, step trace and evidence trail.

Prototype Performance report/Benchmarking

Measure	Result (live Gemini 3.8 Flash, prototype)
End-to-end run, 4 documents, cold	≈ 60 s wall time (reads in parallel); 25–40 s for 1–3 documents
Per step latency	READ 4–20 s per document · VERIFY ≈ 9 s · SCORE < 5 ms · EXPLAIN ≈ 6 s · WRITE ≈ 14 s; repeated runs < 25 s (document reads cached)
Extraction accuracy on synthetic fixtures	12/12 months exact per document type (Pillow-rendered ledger photo, bank CSV, bank PDF, payment export) on three invented businesses; hackathon rules: no real data; not yet measured on real customer documents
Schema validation	100% of model calls validated on the first attempt in test runs
Number guardrail	≈ 100 numbers checked per memo, 0 invented figures in test runs
Verification	18 checks per assessment; declared-vs-bank gap flagged (−3 points) · thin file flagged
Tests	Unit tests for the scorer, verification math, uploads and storage pass
Scorecard calibration	Not yet calibrated: Kervan v0.1 weights, PD estimates and rate bands are expert-set initial policy, pinned by tests, to be recalibrated on partner-bank portfolio data

Additional Details/Future Development (if any)

Next 90 days

- Firebase Authentication for business and bank logins (ID tokens verified in FastAPI) and Identity-Aware Proxy in front of the bank view; Cloud Storage with signed URLs for uploaded documents.
- Registration and tax documents to fully verify tenure and identity; e-signature consent for data sharing.
- Scorecard calibration on portfolio data with a partner bank; monitoring of grade distributions and overrides; step traces to Cloud Logging.
- Gemini through Vertex AI so data stays in-region; Cloud Build trigger on main running the tests before each deploy; bank-side API into loan-origination systems.

Scale story

- Same pipeline for Kazakhstan, Uzbekistan and Kyrgyzstan: a per-market scorecard config and prompt language notes; schemas, verification math, guardrails and deployment stay the same.
- Owners carry their passport between lenders and refresh it as records grow — the credit file follows the business, not the bank.

Provide links to your:

- GitHub Public Repository
- Demo Video Link (3 Minutes)
- Final Product Link

GitHub public repository

https://github.com/taivnaaiwnl/singsing

Demo video (3–4 minutes)

(demo video link)

Final product link (Cloud Run)

(Cloud Run URL)

M bank

Kervan Credit Passport

Gemini reads, verifies, explains and writes. The score comes only from the scorecard.

Team TL · Taivanbat Bayarsaikhan · Lkhagvadorj Munkhdalai

AI Engineers & Data Scientists at M bank, Mongolia's first neobank

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

